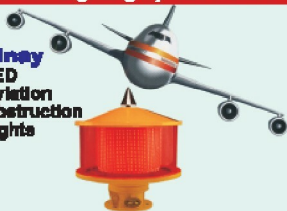


BINAY—Lighting Up The Future**Binay
LED
Aviation
Obstruction
Lights****BINAY LED-based MEDIUM INTENSITY
Aviation Obstruction Light Beacon**

*As per the International Civil Aviation Organization (ICAO) requirements
for use as obstruction above 45m*

*Available in Low Intensity and Medium Intensity versions (as per the
International Civil Aviation Organization guidelines), BINAY's
patented LED Aviation Lights come with a 5-year warranty*

LED Obstruction Lighting for:

- Industrial chimneys and smokestacks
- Transmission, microwave, and cellular towers
- Radio, TV and similar structural towers
- High-rise buildings and structures
- Airports and airfields

The BINAY LED Aviation Obstruction Light offers the following advantages:

- Fit-and-forget maintenance-free operation
- A long life of 100,000 hours (20 years at 12 hours daily burning)
- Runs for itself within a short period of operation in the form of reduced installation, maintenance and servicing costs
- Quick installation; Reliable operation 365 days per year
- Shock-proof and vibration-resistant
- Over-Engineered internally to allow for natural LED internally degradation over its operating lifetime



**THE BINAY LED OBSTRUCTION LIGHT IS UNDER ACCEPTED
PATENT, AND AS SUCH IS A PROPRIETARY PRODUCT**



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and recover quickly—according to UK-based Peratech that has designed the device.

Quantum tunnelling composite material is also used to add touch controls to mobile phones and other devices.

Device invisible to magnetic fields created

Using superconductor and ferromagnetic materials, European researchers have developed a device that hides its contents to make them invisible to a static magnetic field.

Researchers at University of Barcelona, led by Alvar Sanchez, lecturer in the Department of Physics, put forward the mathematical formula to design the device. The device features a cylinder with two concentric layers.

"While the inner layer consists of a superconducting material that repels magnetic fields, the outer layer is a ferromagnetic material that attracts them. Placed in a magnetic field, the device has no effect on the field lines, showing neither a shadow, nor a reflection. So an object inside the device cannot be detected," AFP said in a report.

According to the researchers, the device represents a step towards the invisibility of light. "There are many applications, for some cars, ships or a submarine," Sanchez told AFP. The device could also be used for patients who have a pacemaker and need an MRI of a knee or other body part, so the image would not be distorted.

Ultra-thin electronic films with greater capacity

The development of a new combination of polymers has made it possible to design ultra-thin films capable of self-organisation. The films would be so thin that they would open up new avenues in flexible electronics. These films would also make it possible to increase the capacity of hard disks and the speeds of microprocessors.

A French-American team is developing the combination of polymers by associating sugar with oil-based macromolecules. So far thin films for electronic circuits were developed using synthetic polymers, which were derived from petroleum. The flip side of these films was that their minimum structural resolution was around 20 nanometres and could not be reduced further using petroleum-derived polymers. This created obstacles in developing very-high-resolution flexible electronic devices.

However, the new hybrid material combines sugar-based and petroleum-derived (silicon containing polystyrene) polymers with different physical and chemical characteristics. This copolymer, formed of highly incompatible elementary building blocks, is similar to an oil bubble attached to a small water bubble. The researchers have shown that this type of structure is capable of organising itself into sugar cylinders within a petroleum-based polymer lattice.

Bacteria to help make better nanoelectronics

Researchers at Britain's University of Leeds, in collaboration with Japan's Tokyo University of Agriculture and Technology, have made use of a type of bacterium—Magnetospirillum